Assignment: N-Queens Problem **[Max. Marks: 30]**

Objective

The objective of this assignment is to understand and implement a constraint-based problem using Prolog, by solving the classic N-Queens problem. This assignment demonstrates how declarative programming, recursion, backtracking, and logical constraints can be used to solve combinatorial search problems.

Problem Statement

The N-Queens problem is defined as follows:

Place N queens on an N × N chessboard such that no two queens attack each other.

A queen can attack another queen if they are:

- In the same row
- In the same column
- On the same diagonal

Solution reference

Refer to the following link, which explains the solution using three approaches

https://www.youtube.com/watch?v=l_tbL9RjFdo&list=PLEvH6T-1oh76ndUiFUDjIyrrL9FyrSKXw&index=5

Implement all the three approaches of generating solutions.